Set 1: MACHINE HEALTH & MAINTENANCE

Q1: EXCEL
Create a “Machine Health Score” workbook: per MachineID, compute hourly averages of Temperature, Vibration, Pressure, then flag machines that exceed plant-level mean + 2×std for any metric.

Q2: EXCEL
Create a maintenance planning sheet: calculate daily MaintenanceFlag count per Plant, identify top 10 machines with highest maintenance occurrences, and build a 7-day moving average trend of daily maintenance events.

Q3: SQL
Write queries to return: (a) daily average Temperature, Vibration, Pressure by Plant, and (b) top 5 MachineID per Plant by total DefectCount.

Q4: SQL
Create a view machine_risk_2025 that assigns RiskLevel per MachineID using thresholds derived from dataset percentiles: High risk if Vibration >= 90th percentile OR Temperature >= 90th percentile OR DefectCount >= 90th percentile (else Medium if >= 75th any, else Low).

Q5: PYTHON
Build a regression model to predict DefectCount using Temperature, Vibration, Pressure, EnergyConsumption, ProductionUnits, plus encoded Plant and MachineID. Report MAE and RMSE.

Q6: PYTHON
Build a classification model to predict MaintenanceFlag. Handle class imbalance, report precision/recall/F1, and output top 20 highest-risk future rows (highest predicted probability).

Q7: POWER BI
Create a maintenance dashboard: KPI cards (Total Defects, Avg Vibration, Maintenance Events), line chart of daily maintenance events, and table of top-risk machines (by defects + vibration).

Q8: POWER BI
Create drill-down: Plant → MachineID → hourly sensor trends (Temperature, Vibration, Pressure) with anomaly highlighting using measure vs plant average.